

Python + ML Camp

Sciences Po Paris
Spring 2025

Instructor:	Emir Tarık Dakin	Time:	January 23rd-25th, 9:00-16:00
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Office Hours: You can book an appointment via email.

Course Description: This intensive three-day camp introduces fundamental Python programming and machine learning techniques essential for data analysis in economics and econometrics. The course aims to transition your existing R knowledge to Python, focusing on practical applications in econometric modeling and machine learning. Through interactive sessions and a hands-on project, you will learn to use Python's powerful libraries to analyze real-world data effectively.

By the end of this camp, you should have a solid understanding of Python syntax, data manipulation with Pandas, data visualization, and the ability to implement basic econometric and machine learning models using Scikit-Learn. You will also gain experience in handling a complete data science project, from data preprocessing and feature engineering to model evaluation and optimization.

Objectives: The primary aim is to equip you with practical Python programming and machine learning skills applicable in economics, policy analysis, and data science fields.

Attendance: Attendance is crucial for this intensive camp due to the condensed schedule and cumulative nature of the material. Please ensure you attend all sessions and actively participate in discussions and exercises. Bringing your laptops to class is highly recommended as there will be hands-on coding activities throughout the camp.

Prerequisites: Familiarity with basic coding principles and experience with R is advised. A fundamental understanding of statistics and econometrics is required. No prior knowledge of Python is assumed.

Software Requirements: We will be using Python extensively throughout this camp, focusing on libraries such as Pandas, Matplotlib, Seaborn, and Scikit-Learn. For collaborative purposes and ease-of-use, we will mainly use Google's [Colab](#).

To get started with Python, you can refer to the [LearnPython.org](https://learnpython.org) tutorials for basic syntax and concepts. Additional resources and materials will be provided during the camp.

Project Overview: As a significant component of this camp, you will undertake an individual project where you will select a research question of your own choosing. You will be responsible for finding a suitable dataset, applying the data analysis and machine learning techniques learned during the camp to analyze your question, and interpreting your results. This project will culminate in a final presentation and a detailed report,

allowing you to demonstrate your understanding and practical application of the concepts covered.

Grading Policy: Your project work will determine your course grade. There will be no exams, quizzes or homework.

- **Presentations (60%)**

You will present your work in short sessions (about 10 minutes) which will be followed by questions. There will be three presentation sessions (worth 20% each) for each group. See the Thursday sessions for further details. Grades will be based on the quality of the presentation, your oral defense and ability to answer questions.

- **Report (40%)**

At the end of the semester, you will turn in a short replica of an academic paper, which will include your story, data/methodology and your findings. Grades will be based on the quality of the writing of the paper, the justification for the model you estimate, and the accuracy with which you interpret your statistical models.

Day 1: Introduction to Python and Data Handling

Morning Session

9:00 AM – 10:30 AM: Introduction to Python syntax

- **Variables and data types:** Strings, integers, floats, booleans.
- **Basic operations:** Arithmetic operations, string manipulation.
- **Control structures:** if-else statements, for and while loops.
- **Functions:** Defining and calling functions, scope of variables.

10:30 AM – 10:45 AM: Break

10:45 AM – 12:00 PM: Working with data structures

- **Lists, Tuples, and Dictionaries:** Creation, manipulation, and iteration.
- **List comprehensions:** Efficient ways to create lists.
- **Modules and Libraries:** Importing and using standard libraries.

Afternoon Session

1:00 PM – 2:30 PM: Data handling with Pandas

- **Introduction to Pandas DataFrames and Series**
- **Reading and writing data:** CSV, Excel files.
- **Data cleaning:** Handling missing values, duplicates.
- **Data exploration:** Descriptive statistics, data slicing.
- **Interactive exercise:** Load and explore the Titanic dataset; identify and handle missing values.

2:30 PM – 2:45 PM: Break

2:45 PM – 4:00 PM: Data visualization with Matplotlib and Seaborn

- **Basic plots:** Line, bar, histogram, scatter plots.
- **Customization:** Titles, labels, legends.
- **Seaborn for statistical plots:** Regression plots, heatmaps.
- **Interactive exercise:** Visualize the relationship between passenger age and survival rate in the Titanic dataset.

Day 2: Econometric Models with Scikit-Learn

Morning Session

9:00 AM – 10:30 AM: Advanced Python concepts

- **Exception handling:** try, except blocks.
- **File I/O:** Reading from and writing to files.
- **Lambda functions and map/filter/reduce**
- **Interactive exercise:** Write a program that reads a text file, counts the frequency of each word, and handles file exceptions gracefully.

10:30 AM – 10:45 AM: Break

10:45 AM – 12:00 PM: Introduction to Scikit-Learn

- **Understanding machine learning workflows**
- **Data preprocessing:** Train-test split, scaling features.
- **Linear regression:**
 - Brief theory recap of linear regression.
 - Implementing using Scikit-Learn.
 - Interpreting coefficients.
- **Interactive exercise:** Apply linear regression to predict house prices using a sample dataset.

Afternoon Session

1:00 PM – 2:30 PM: Econometric models

- **Multiple linear regression**
- **Logistic regression:**
 - Binary classification problems.
 - Predicting probabilities.
- **Evaluation metrics:**
 - Regression: MSE, RMSE, MAE, R^2 .
 - Classification: Accuracy, Precision, Recall, F1 Score.
- **Interactive exercise:** Evaluate a logistic regression model predicting Titanic passenger survival.

2:30 PM – 2:45 PM: Break

2:45 PM – 4:00 PM: Feature engineering and selection

- **Handling categorical variables:** One-hot encoding.

- **Feature scaling:** Standardization and normalization.
- **Feature selection techniques:** Correlation matrices, LASSO.
- **Interactive exercise:** Improve the predictive model by engineering new features from the Titanic dataset.

Day 3: Machine Learning Techniques

Morning Session

9:00 AM – 10:30 AM: Ensemble methods

- **Decision trees:**
 - Understanding tree-based models.
 - Overfitting and pruning.
- **Random forests:**
 - Concept of ensembling.
 - Feature importance.
- **Interactive exercise:** Implement decision tree and random forest models on the Titanic dataset and compare results.

10:30 AM – 10:45 AM: Break

10:45 AM – 12:00 PM: Model optimization

- **Cross-Validation techniques:**
 - K-Fold cross-validation.
 - Leave-one-out cross-validation.
- **Hyperparameter tuning:**
 - Grid Search with `GridSearchCV`.
 - Randomized Search.
- **Interactive exercise:** Optimize the random forest model using cross-validation and hyperparameter tuning.

Afternoon Session

1:00 PM – 2:30 PM: Introduction to unsupervised learning

- **Clustering techniques:**
 - K-Means Clustering.
 - Hierarchical Clustering.
- **Principal Component Analysis (PCA):**

- Dimensionality reduction.
- Visualizing high-dimensional data.
- **Interactive exercise:** Apply PCA and clustering to the Iris dataset to identify species clusters.

2:30 PM – 2:45 PM: Break

2:45 PM – 4:00 PM: Project planning and next steps

- Begin working on selected projects.
- Q&A session and establishing goals.
- **Wrap-Up:** Summary of key concepts and expectations moving forward.

Thursday sessions: Project work

Session 1: Thursday, 20/2, 7:15 PM – 9:15 PM

- **Progress presentations (~10 mins):** Each student/group presents their project progress and makes an introduction. Talking about the context, the literature and the data.
- Q&A and feedback for the next session.

Session 2: Thursday, 27/3, 7:15 PM – 9:15 PM

- **Progress presentations (~10 mins):** Students present their selected methodology, followed by initial findings. Discussions on improving the methodology.
- Students must complete their project until the next session.

Session 3: Thursday, 25/4, 7:15 PM – 9:15 PM

- **Final presentations (~10 mins):** Students deliver comprehensive presentations of their projects.
- Detailed report to be delivered by **May 9th**.